

# SIADS 696 Milestone II Project Report

## Frequent Mental Distress Prediction Using Air Pollution

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### Introduction

The motivation behind this project aims to utilize previous findings from related literature and leverage supervised and unsupervised learning to answer two questions:

1. Can a county's air quality and profile reliably predict its mental health needs next year?
2. Do high pollution or high distress clusters share common geographic or industrial traits?

Note that we do not aim to claim causality; we are conducting a predictive and exploratory exercise.

### *What problem are you trying to solve?*

It's well-known beyond anecdote that health outcomes are influenced by many environmental, economic, and social factors; yet, it remains unclear whether on a population-level (here, using county-level characteristics), if we can use data to anticipate future community mental health needs. Our project investigates whether annual air quality, climate, and socioeconomic conditions can predict next-year **Frequent Mental Distress (FMD)**, defined by the **CDC PLACES** dataset as the percentage of adults reporting poor mental health for 14 or more days in the past month. As part of our project, we also explore if counties naturally group/cluster into distinct environmental and socioeconomic profiles along differing mental health outcomes. The problem we are trying to solve is if it's at all possible to predict mental health distress at a county-level using pollution data as an informative tool.

### *What impact will solving the problem have?*

If community mental health needs can be predicted even modestly, public health agencies could better allocate resources and identify at-risk regions pre-emptively. Understanding the limits of county-level data helps identify what additional information may be needed for reliable forecasting. Our work also provides an integrated county-year dataset that can support future research on environmental determinants of mental health.

### *What motivated you to work on it?*

A growing body of literature suggests that air pollution, climate, and socioeconomic conditions are associated with mental health outcomes, but relatively few studies have examined whether these variables can predict future community-level mental distress using machine learning. We were interested in evaluating whether publicly available environmental and demographic data contain enough predictive signal to forecast next-year mental health and whether unsupervised learning could reveal meaningful county profiles beyond traditional statistical analyses.

### *Summarize your project's supervised and unsupervised methods, highlighting any novel contributions.*

As such, the primary objective of this project was to construct a predictive framework based on a county's current socio-environmental and economic ecosystem. Specifically, the target variable was defined as **Next-Year Frequent Mental Distress (FMD)** prevalence, which is the percentage of adults in a given county reporting 14 or more mentally unhealthy days per month, measured one year following the observed features.

We constructed an enriched county-year panel containing **14,748 observations across 3,008 U.S. counties and county-equivalents from 2019–2023** by combining data from CDC PLACES, EPA

AirData, NOAA climate records, the American Community Survey (ACS), and Census county geographies.

Predicting public health outcomes within a spatial panel structure introduces a critical methodological challenge: temporal data leakage. To address this, our supervised learning design employed a leakage-safe forecasting framework in which features from year  $t$  were used to predict FMD in year  $t + 1$ . We evaluated multiple regression models, including Elastic Net, Random Forest, Gradient Boosting, and XGBoost, using cross-validation and a held-out test set.

For the unsupervised learning component, we standardized all non-FMD variables and applied DBSCAN clustering to a low-dimensional embedding. FMD was intentionally excluded during clustering and reintroduced afterward to compare mental health outcomes across the discovered clusters.

| Research question                                              | Modeling approach                                  | Interpretation target                                     |
|----------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------|
| Can current-year county conditions predict next-year FMD?      | Supervised regression with year-shifted target     | Predictive usefulness and error patterns                  |
| Do counties form meaningful socio-environmental profiles?      | UMAP/PCA-style embedding plus DBSCAN               | Cluster differences in held-out FMD and profile variables |
| Are high-FMD areas geographically or economically distinctive? | Residual maps, cluster maps, and profile summaries | Exploratory evidence for follow-up study                  |

The best supervised model was a tuned Random Forest, achieving a holdout RMSE of 1.98 and MAE of 1.65, compared with the Dummy Regressor baseline RMSE of 2.43. Holdout  $R^2$  remained negative ( $-0.02$ ), indicating that county-level environmental, climate, and socioeconomic variables alone do not provide sufficient predictive signal to reliably forecast next-year FMD. At this point, it would not be reasonable to claim that we can reliably predict mental health distress with the current feature set.

The unsupervised analysis revealed one dominant cluster along with several smaller outlier clusters. Because FMD was excluded from clustering and examined only afterward, differences in FMD across clusters can be interpreted as an independent outcome comparison. While the clustering did not reveal clear, balanced county profiles, it identified meaningful environmental and socioeconomic groupings that may serve as a useful starting point for future hypothesis generation and targeted public health investigations. We conclude that while the existing body of county-level data is currently limited for predicting future FMD, it remains a valuable starting point for future exploratory profiling.

## Related Work

There is an extensive and rapidly growing body of research dedicated to the environmental determinants of mental health at the community and county levels in the United States. A lot of research revolves around the impact of environmental factors on mental illness risk. This is true for “Particulate matters (PM2.5, PM10) and the risk of depression among middle-aged and older population. Analysis of the Korean Longitudinal Study of Aging (KLoSA), 2016–2020 in South Korea” ([Park, et al.](#)) found that prolonged exposure of particulate matters increased risk of depression in the elderly. Additionally, “The impact of neighborhood environment on mental health: Evidence from China” ([Lei, et al.](#)) concluded that air pollution perception, coupled with socioeconomic status, was crucial for mental health in Chinese communities. Although these studies focus on age-based cohorts in Asian communities, our research will focus solely at population level across U.S. counties to model a possible correlation between air quality and mental health. The Gulf Long-Term Follow-up Study ([Werder et al., 2024](#)), explored how PM2.5 and greenspace impact depression in the Southeastern U.S., pointing to potentially similar vulnerabilities domestically.

Our project extends this prior work in two key ways. First, rather than focusing on specific age cohorts or geographic regions, we model county-level outcomes across the entire contiguous United States. Second, we incorporate unsupervised learning to uncover latent socio-environmental archetypes across U.S. counties. This project also builds on our own prior Milestone I work, which initially explored the relationship between air quality and health outcomes and motivated the expanded scope of the current analysis.

## Data Sources

This project utilizes data from five primary sources: AirData, CDC PLACES, and US Census (ACS) Data. The modeling panel is county-year and uses five-digit county GEOID as the primary merge key. The original imputed panel was extended with additional ACS, EPA, NOAA, and Census geography files to satisfy the project requirement for socioeconomic controls, daily exposure features, climate profile variables, industrial traits, and spatial visualization.

| Source             | Format              | Key Variables                                                                             | Coverage; role                                                   | Records         |
|--------------------|---------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------|
| CDC PLACES         | CSV                 | FMD prevalence, lower and upper confidence intervals                                      | 2018–2023; Mental health outcome by county-year                  | ~18K county-yr  |
| EPA AirData Annual | CSV                 | Days with AQI, good/moderate/unhealthy days, Max AQI, Median AQI, pollutant-defining days | 2019–2023; Annual air quality predictors                         | ~6.5K county-yr |
| EPA AirData Daily  | ZIP/CSV             | Daily AQI mean, standard deviation, p90, max, days >100, >150, >200                       | 2019–2023; Exposure variability and extreme-event features       | ~5.4M daily     |
| ACS 5-Year         | Pipe-delimited .dat | Income, poverty, unemployment, education, insurance, industry composition                 | 2019–2023; Socioeconomic, health-access, and industrial controls | ~3.1K × 5       |
| NOAA Climate       | JSON                | Annual average temperature, precipitation, historical anomalies                           | 2019–2023; County climate profile                                | ~3.1K × 5       |
| Census Boundaries  | Shapefile ZIP       | County polygon geometries                                                                 | 2023; Choropleth and spatial visualization support               | 3,234 polygons  |

*Table 1. List of datasets used for analysis.*

Our enriched dataset contains 83 columns. ACS and NOAA features cover nearly all rows. EPA daily AQI features cover only counties with daily monitor reporting, so many missing daily features are imputed.

Because of limitations in the years of data available across the main datasets (PLACES being a key bottleneck), in the harmonization step we had to accept losing granularity and also years. Although it was originally derived from BRFSS, PLACES was not a simple feature engineering downstream derivative dataset where we could have stitched together everything from scratch ourselves for the scope of the project.

## Feature Engineering

First, since we were analyzing data from multiple datasets, we had to merge them into a single table. Fortunately, the measurements of each dataset were all taken at county-level granularity, so merging them by state and county name was straightforward. After doing so, we noticed a lot of rows were missing environmental measurements from AirData. For this imputation, we used a BallTree to perform a K-nearest-neighbor algorithm to fill in missing values. Essentially, we took the geographically nearest 10 points with valid AirData measurements, and weighed them by distance to the null point in question.

Now, with our full, merged dataset, we could perform some feature engineering. First, we converted some of our scalar columns (population, number of unemployed people, etc.) into ratios (e.g. unemployment rate). This helps to normalize our dataset and to encourage intensity-analysis rather than scale. Afterwards, the previous columns used to compute these ratios were dropped to prevent multicollinearity.

Finally, for each county, we shifted FMD forward one year, so that current-year features predict next-year FMD. We did this to address data leakage concerns we had during the early stages of this analysis. The final output dataset was distilled to 14,748 records across 24 columns. Features in this dataset include, in addition to FMD, environmental air quality and climate data, as well as socioeconomic, educational, and industry demographic data. A full list of these features is provided in **Appendix B**.

## Supervised Learning

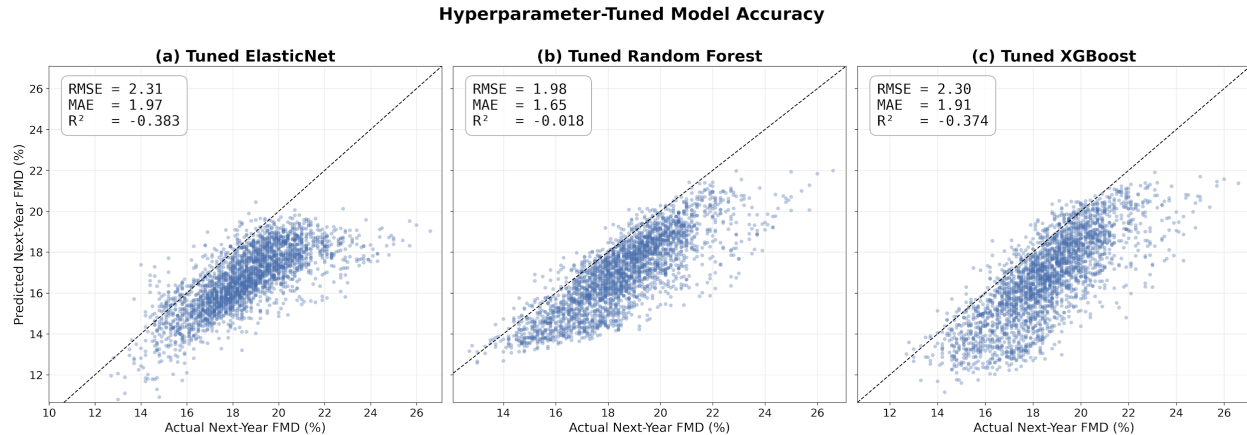
The supervised task was primarily a regression problem: predict next-year FMD prevalence. As a baseline, we chose a Dummy Regressor that always predicts the mean FMD to establish a minimum performance threshold our other models should outperform.

For our supervised learning models, we chose ElasticNet because some groups of features, such as environmental and socioeconomic ones, could exhibit multicollinearity (e.g., poverty rate correlates with uninsured rate), and ElasticNet accounts for this through combined L1 and L2 regularization while performing implicit feature selection. As a linear model, it also establishes whether the relationship between features and FMD is approximately linear. We also decided to train a Random Forest model due to its ability to model non-linear relationships. For example, the effect of air quality may differ by income level. We also benefit from being able to examine feature importances through it. Finally, Gradient Boosting (XGBoost) was selected as a sequential ensemble method that iteratively corrects residual errors, complementing Random Forest's parallel averaging approach. This gives us two ensemble methods for comparison: bagging vs. boosting.

To improve each model's performance, we performed hyperparameter tuning using **GridSearchCV** with 5-fold cross-validation on the training dataset. Instead of using just a single initial configuration to move forward, we explored several parameter settings for each model. For **Random Forest**, we searched over `n_estimators`  $\in \{100, 250\}$ , `max_features`  $\in \{\text{sqrt}, 0.5, 0.7, 1.0\}$ , and `min_samples_leaf`  $\in \{1, 3, 8, 15\}$ . For **Gradient Boosting**, we searched over `n_estimators`  $\in \{100, 250\}$ , `max_depth`  $\in \{3, 5\}$ , and `learning_rate`  $\in \{0.01, 0.1\}$ . For **ElasticNet**, we searched over `alpha`  $\in \{0.01, 0.1, 1.0\}$  and `l1_ratio`  $\in \{0.1, 0.5, 0.9\}$ . The configuration with the lowest mean cross-validation RMSE was selected for final holdout evaluation. This approach allowed us to compare multiple model configurations while balancing predictive accuracy with generalization to unseen data.

## Evaluation

Since this is a regression problem, we used RMSE, MAE, and  $R^2$  as our evaluation metrics, along with grouped cross-validation by year where applicable. We primarily chose RMSE because large errors are undesirable in a sensitive metric like FMD. MAE provides a complementary metric that is not sensitive to outliers, and  $R^2$  allows us to compare how much variance is explained relative to our baseline model.



*Figure 1. Hyperparameter-tuned model accuracy: (a) tuned ElasticNet model, (b) tuned Random Forest model, and (c) tuned XGBoost model. Random Forest achieved the lowest holdout RMSE (1.98), suggesting that nonlinear tree-based methods captured somewhat more county-level variation than ElasticNet, although all models remained relatively close to the Dummy baseline.*

| Model                   | Holdout RMSE | Holdout MAE | Holdout $R^2$ | CV RMSE (mean $\pm$ SD) |
|-------------------------|--------------|-------------|---------------|-------------------------|
| Random Forest tuned     | 1.98         | 1.65        | -0.02         | 2.02 $\pm$ 0.63         |
| Gradient Boosting tuned | 2.30         | 1.91        | -0.37         | 2.16 $\pm$ 0.49         |
| ElasticNet tuned        | 2.31         | 1.97        | -0.38         | 2.32 $\pm$ 0.59         |
| Dummy mean baseline     | 2.43         | 1.95        | -0.54         | 2.73 $\pm$ 0.46         |

*Table 2. Table and comparison of evaluation metrics for dummy and supervised learning models. CV metrics report mean  $\pm$  SD across 5 folds.*

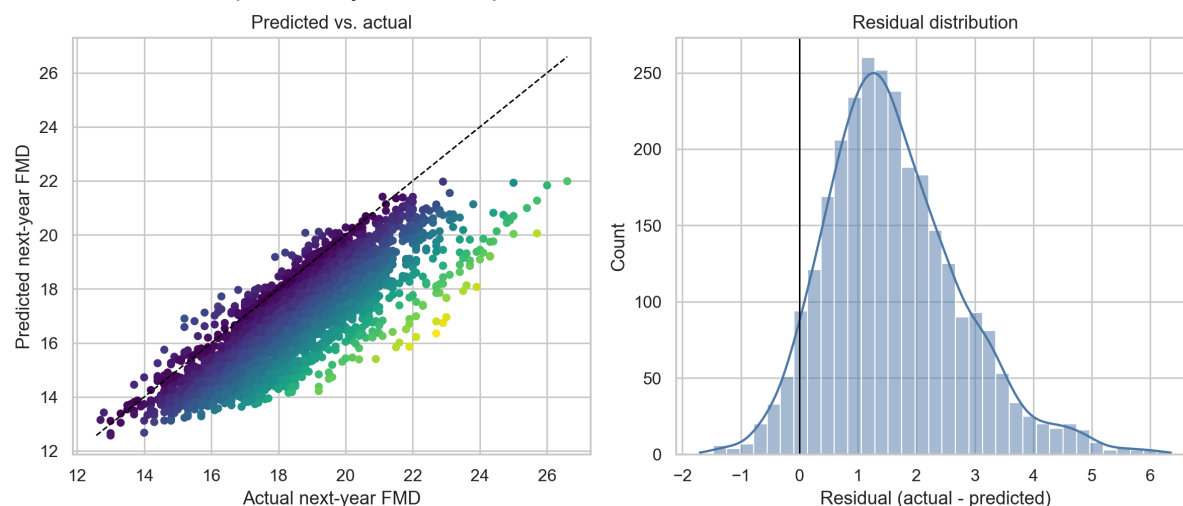
## Findings

Referring to our results in **Table 2**, the tuned Random Forest achieved the lowest holdout RMSE (**1.98**) compared with the Dummy mean baseline (**2.43**). The improvement of **0.45 RMSE** is simply the difference between these two values: **2.43 – 1.98 = 0.45 RMSE**.

This reduction suggests the model captures some predictive signal beyond the baseline. However, the negative holdout  $R^2$  value (**-0.02**) indicates it does not explain variation in the holdout data better than a simple mean predictor under strict temporal split. Taken together, these results suggest a modest improvement over the baseline rather than strong forecasting.

Our residual analysis provides one of the more informative supervised results. It shows that the model tended to underpredict counties with the highest observed FMD. In other words, the largest prediction errors occurred in counties experiencing the greatest mental distress. This pattern points to the possibility

that important county-level factors (e.g., healthcare access, disability prevalence, greenspace, disaster exposure, or behavioral health infrastructure) are not adequately represented in the available annual datasets. At the same time, we want to be careful not to lean too heavily on any one suspected missing variable. Instead, we interpret this result as simply: county-level averages can erase granularity and extreme outcomes, particularly in underrepresented communities.



*Figure 2. Predicted vs. actual next-year FMD and residual distribution on the 2022→2023 holdout. We see predictions show a moderate positive relationship with observed FMD but are compressed toward the mean (scatter falls along the 45° line). This actually would point at underprediction of counties with the highest levels of mental distress. The residual distribution indicates systematic prediction error consistent with the model’s negative holdout  $R^2$ . Because the residuals aren’t centered around zero at all, it would make us suspicious of systemic bias working against our current initial efforts.*

## In-Depth Evaluation

### Feature Importance and Random Forest Grouped Ablation

| Removed Family            | Features Removed | Holdout RMSE | RMSE Delta |
|---------------------------|------------------|--------------|------------|
| Education & health access | 5                | 2.06         | +0.08      |
| Industry composition      | 13               | 2.01         | +0.03      |
| Annual AQI                | 15               | 1.99         | +0.01      |
| None (full model)         | 0                | 1.98         | 0.00       |
| Daily AQI exposure        | 25               | 1.98         | -0.00      |
| Socioeconomic scale       | 9                | 1.94         | -0.04      |
| Climate                   | 6                | 1.94         | -0.04      |

*Table 3. Feature-family ablation for random forest results.*

For the chosen model, Random Forest, impurity-based feature importance ranked education and climate variables highly (**Figure 3**). Correlated predictors can share importance, so these rankings should be considered cautiously. We therefore complemented the feature importance analysis with a grouped

ablation analysis, which retrained the model after removing each feature family and measured the resulting change in holdout RMSE.

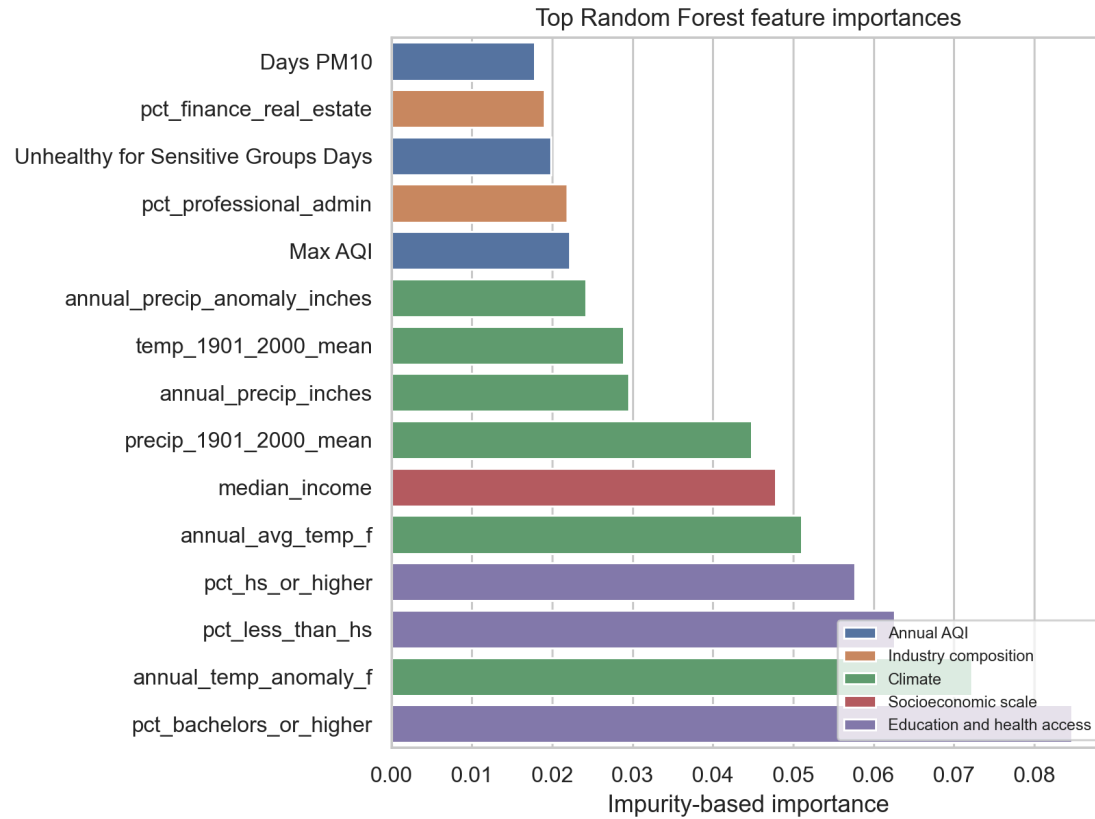


Figure 3. Impurity-based feature importance for Random Forest.

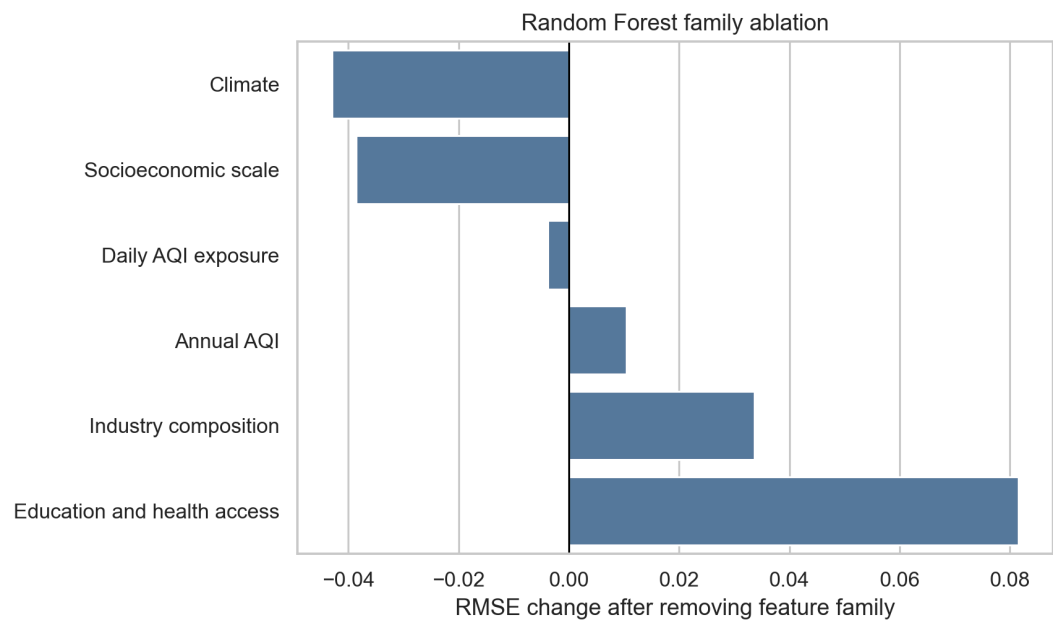


Figure 4. Feature-family ablation for the tuned Random Forest model.

Referring to **Table 3** and **Figure 4**, removing the education and health access feature family increased holdout RMSE from **1.98** to **2.06 (+0.08)**, the largest increase observed across all feature families. Removing the annual AQI feature family increased holdout RMSE by only **+0.01**, while removing the daily AQI feature family produced essentially no change. Given these results, it makes us wonder if education and health access variables may have contributed more consistently to predictive performance than the annual and daily air quality feature families we have now.

### Sensitivity and Tradeoffs

Sensitivity analysis varied Random Forest minimum leaf size and max-feature sampling. The best setting used small leaves and sqrt feature sampling, while larger leaves generally reduced variance but worsened holdout RMSE. The tradeoff is interpretability and stability versus nonlinear flexibility: Random Forest performed best, but at the cost of reduced transparency relative to ElasticNet.

| min_samples_leaf | max_features | Holdout RMSE | Holdout R <sup>2</sup> |
|------------------|--------------|--------------|------------------------|
| 1                | sqrt         | 1.98         | -0.02                  |
| 3                | sqrt         | 2.00         | -0.04                  |
| 8                | sqrt         | 2.02         | -0.06                  |
| 15               | sqrt         | 2.05         | -0.09                  |
| 1                | 0.7          | 2.06         | -0.10                  |
| 1                | 0.5          | 2.06         | -0.10                  |

Table 4. Random Forest sensitivity analysis, top settings. Comparison of different parameters on evaluation metrics.

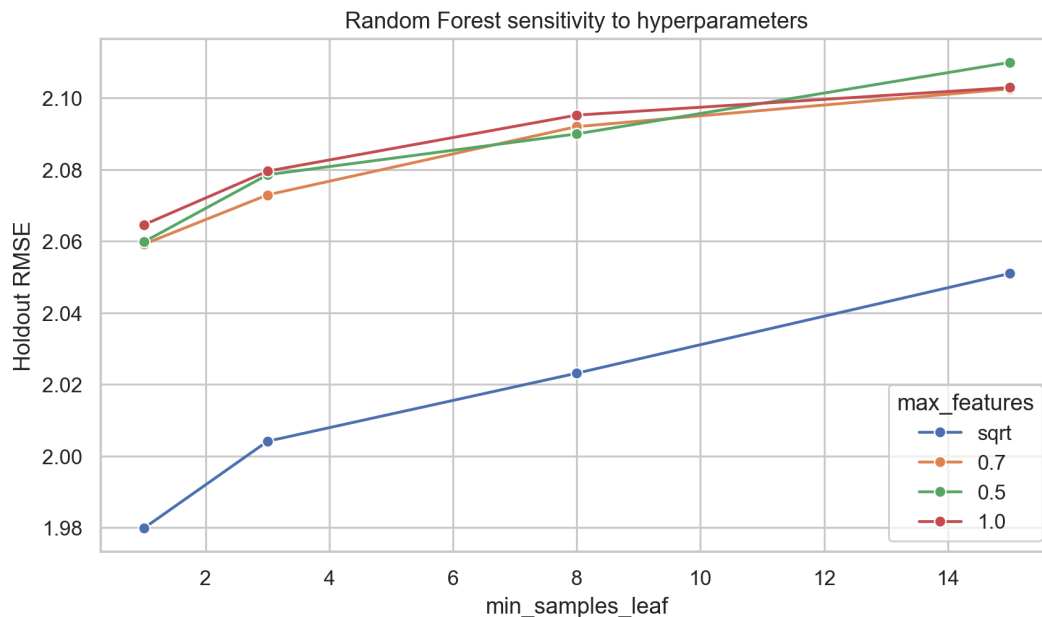


Figure 5. Sensitivity analysis. Smaller leaves with sqrt sampling consistently outperform.

The tradeoff is interpretability and stability vs. nonlinear flexibility: Random Forest performed best, but at the cost of reduced transparency relative to ElasticNet.

Failure Analysis

We focused on examining largest absolute residuals, because we wanted where (in what counties) the model performed worst. The model mainly failed by underpredicting counties in the high-FMD tail, including Appalachian and rural counties where county-level pollution and ACS controls likely miss local behavioral-health infrastructure, disability burden and other hidden social/environmental variables.

Random Forest holdout residuals by county  
positive = actual FMD higher than predicted

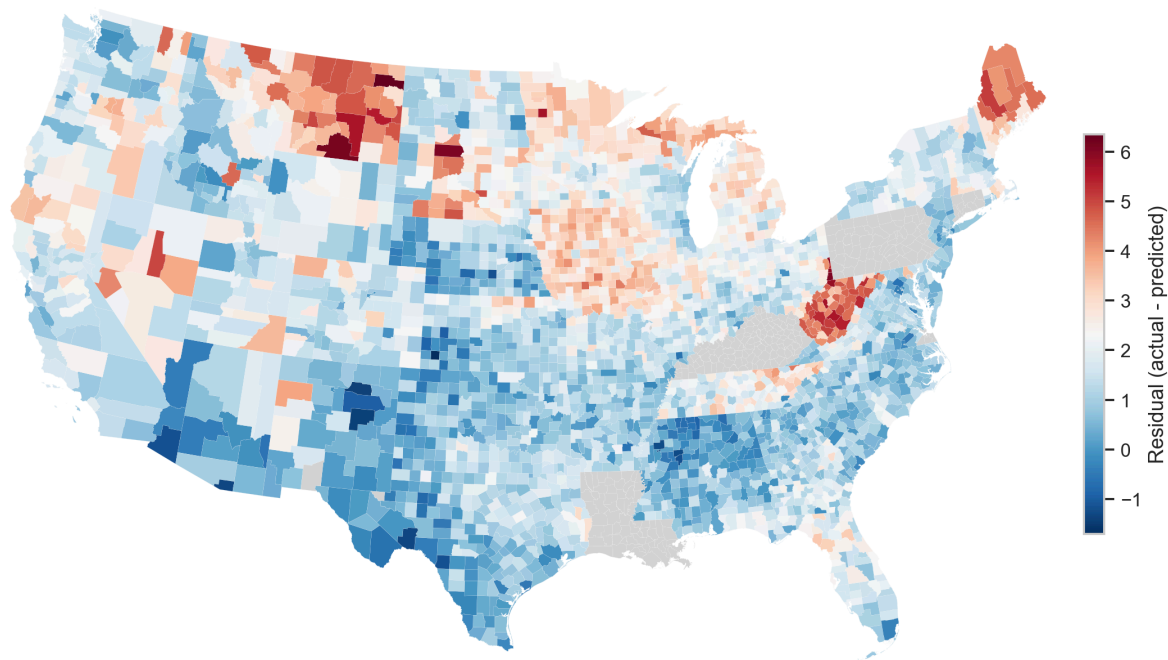


Figure 6. Geographic visualization pattern of supervised residuals distribution. Underprediction concentrates in Appalachia, the Northern Plains, and the rural South. High positive residuals mark counties where actual FMD was higher than expected.

Our failure analysis uncovered three recurring sources for large prediction error:

- 1. **High-distress tail underprediction in rural counties** → *future direction*: incorporate healthcare provider density and disability-related variables.
- 2. **Appalachian underprediction in counties affected by structural health disparities** → *future direction*: incorporate RUCC rurality codes and county-level opioid burden indicators.
- 3. **Underprediction in counties containing tribal lands and Indigenous communities** → *future direction*: integrate IHS service area data and tribal health indicators.

| County    | State | Actual | Predicted | Residual | Category           |
|-----------|-------|--------|-----------|----------|--------------------|
| Roosevelt | MT    | 22.70  | 16.35     | +6.35    | High-distress tail |
| Big Horn  | MT    | 22.90  | 16.74     | +6.16    | High-distress tail |
| Corson    | SD    | 21.90  | 15.86     | +6.04    | High-distress tail |
| Marshall  | WV    | 23.00  | 16.96     | +6.04    | Appalachian        |

|          |    |       |       |       |                                 |
|----------|----|-------|-------|-------|---------------------------------|
|          |    |       |       |       | structural                      |
| Ohio     | WV | 22.10 | 16.22 | +5.88 | Appalachian structural high-FMD |
| Wetzel   | WV | 23.90 | 18.07 | +5.83 | Appalachian structural high-FMD |
| Mahnomen | MN | 21.50 | 15.84 | +5.66 | Indigenous reservation          |
| Rosebud  | MT | 21.60 | 16.01 | +5.59 | Indigenous reservation          |

*Table 5. Failure analysis: identified high-error examples. Largest prediction errors by category Predictions with the highest residuals with actual FMD.*

## Unsupervised Learning

### Methods & Motivation

Our unsupervised analysis asked if counties fall into distinct socio-environmental clusters. We intentionally excluded FMD because we wanted to discover clusters based only on environmental, climate, socioeconomic, healthcare access, education, and industry characteristics, then explore further if those clusters differed in FMD (after). We selected two complementary clustering methods: **DBSCAN** and **K-Means**.

**DBSCAN** was selected because it does not require pre-specifying the number of clusters and can identify outlier counties as noise. **K-Means** was chosen as a complementary centroid-based method that assigns every county to exactly one cluster, making it useful for generating interpretable county archetypes. Before clustering, we standardized numeric county-profile features using **StandardScaler** so variables measured on different scales contributed comparably. The final reproducible report pipeline used **PCA** for the two-dimensional diagnostic embedding and cluster visualizations. As shown in **Figure 11**, we searched **70 DBSCAN parameter combinations** across neighborhood radius (**eps**) and minimum sample (**min\_samples**) values before selecting **eps = 1.43** and **min\_samples = 5**. **Figure 7** shows the K-Means sensitivity search across  $k \in \{2, 3, 4, 5, 6\}$ ; **k = 2** achieved the highest silhouette score (**0.89**) on the PCA embedding.

### Unsupervised Evaluation

We used silhouette scores to evaluate how well counties fit within their assigned clusters. Because K-Means is centroid-based, we also used the Calinski-Harabasz index as a complementary evaluation metric. Since FMD was excluded from clustering, we compared held-out FMD across clusters using one-way ANOVA.

| Method | Selected parameters | Clusters | Noise % | Silhouette embedding | Silhouette scaled features | CH    |
|--------|---------------------|----------|---------|----------------------|----------------------------|-------|
| DBSCAN | eps=1.43, min_s=5   | 2        | 2.2%    | 0.76                 | 0.45                       | —     |
| KMeans | k=2                 | 2        | 0.0%    | 0.89                 | 0.75                       | 321.5 |

*Table 6. Unsupervised method comparison.*

| Method | F statistic | p-value | Groups |
|--------|-------------|---------|--------|
| DBSCAN | 3.65        | 0.056   | 2      |
| KMeans | 8.63        | 0.003   | 2      |

Table 7. Held-out FMD ANOVA across clusters.

### Unsupervised Sensitivity Analysis

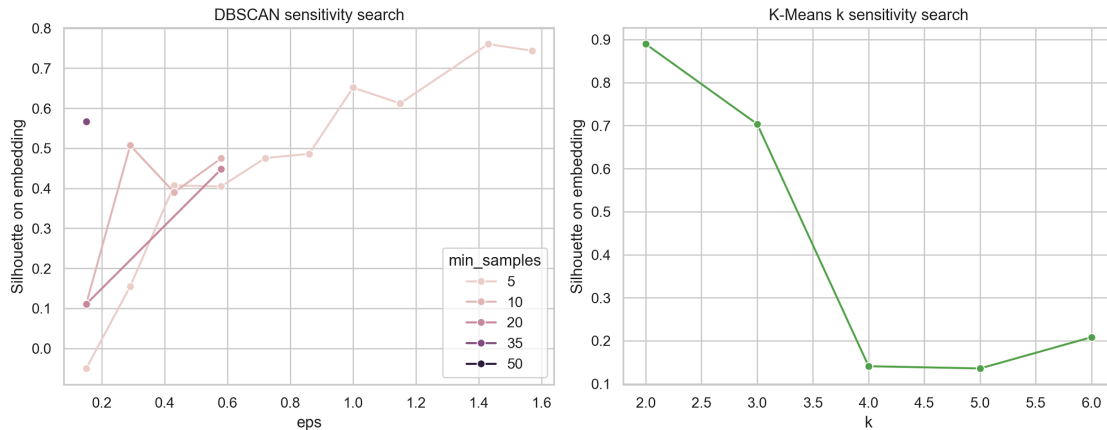


Figure 7. DBSCAN parameter search across 70 configurations. Selected  $\text{eps}=1.43$ ,  $\text{min\_samples}=5$  balances silhouette (0.76) against minimal noise (2.2%).

### DBSCAN Results

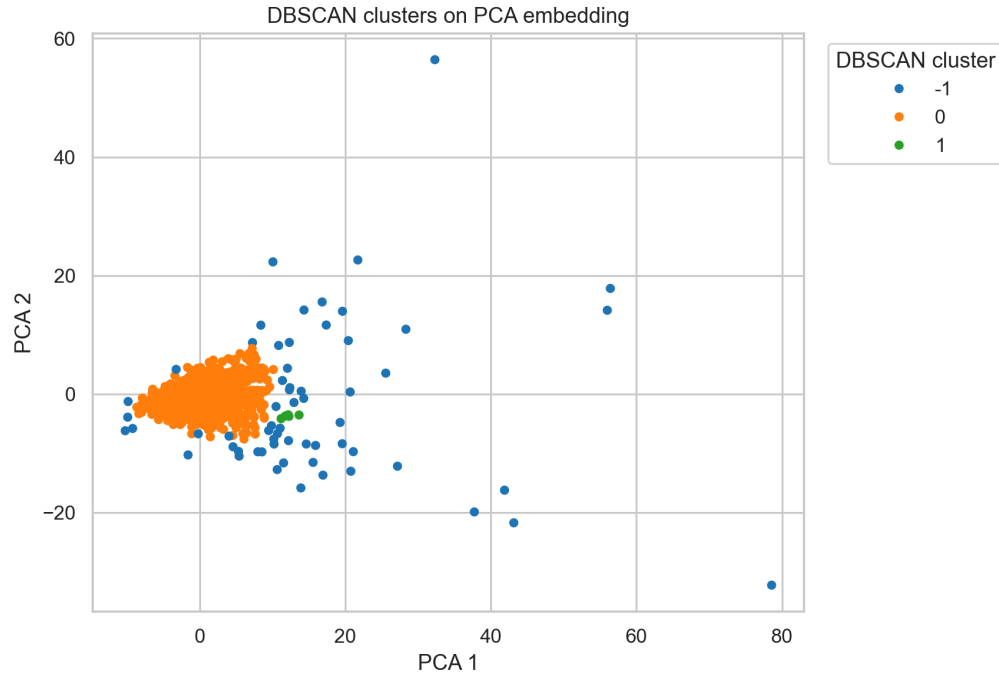


Figure 8. DBSCAN clusters on PCA components. One dominant cluster absorbs most counties, with 61 noise points and a 5-county micro-cluster.

| Cluster    | Counties | Avg FMD | Median AQI | Uninsured | Bachelor+ |
|------------|----------|---------|------------|-----------|-----------|
| -1 (noise) | 61       | 17.06   | 48.8       | 9.6%      | 34.5%     |
| 0 (main)   | 2,746    | 18.61   | 41.4       | 9.5%      | 24.0%     |
| 1 (small)  | 5        | 16.94   | 56.2       | 10.0%     | 39.0%     |

Table 8. DBSCAN cluster profiles.

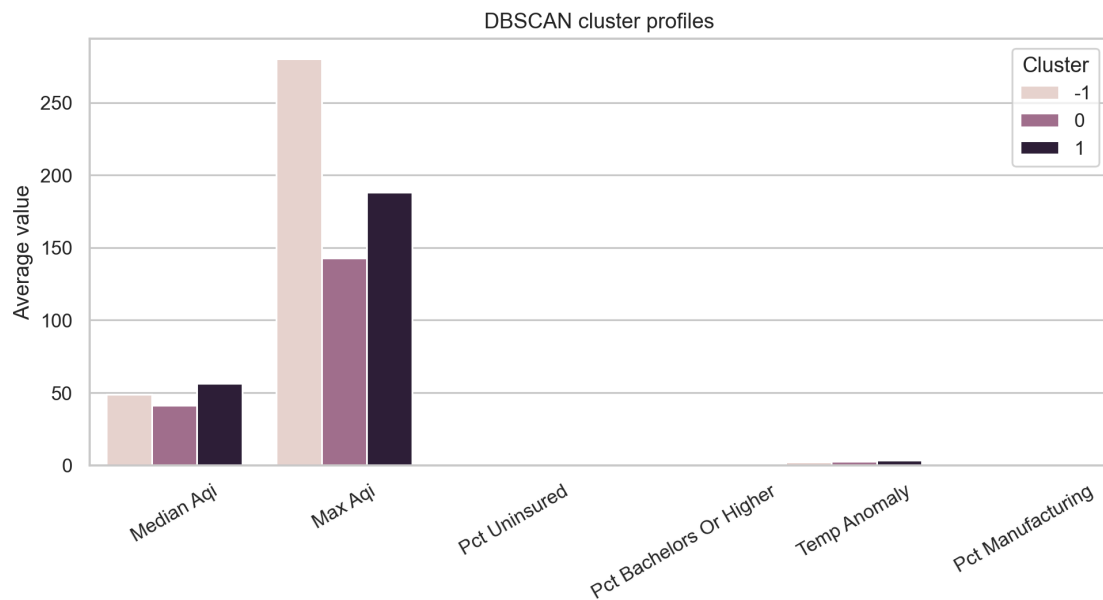


Figure 9. DBSCAN cluster profiles show elevated AQI and higher education; cluster 1 has lower FMD.

### K-Means Results

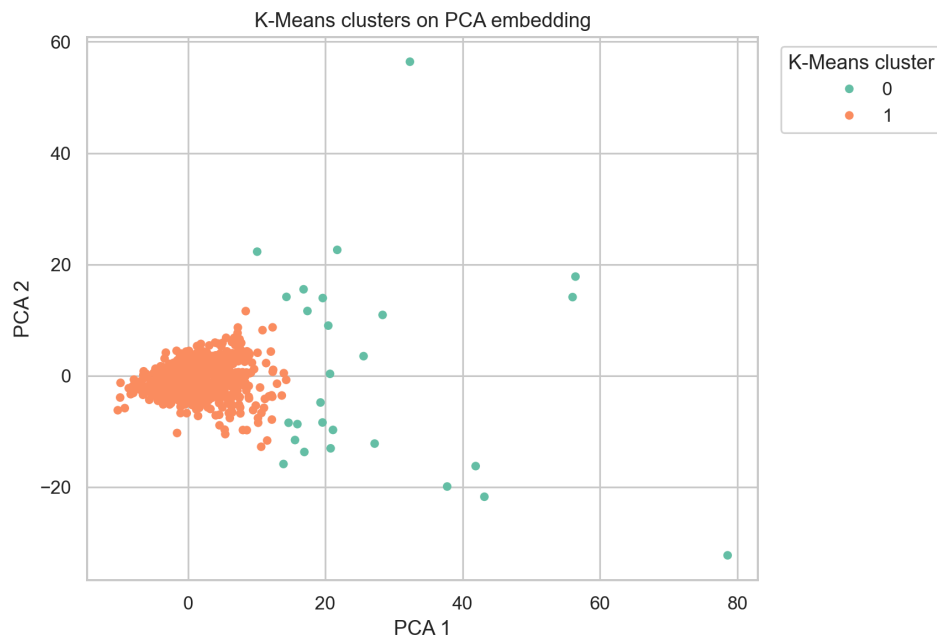


Figure 10. K-Means ( $k=2$ ) on PCA embedding. Dominant cluster contains 2,785 of 2,812 counties.

K-Means produced two clusters with stronger silhouette scores, but one cluster was still much smaller than the other. Specifically, silhouette dropped sharply from  $k=2$  (0.89) to  $k=3$  (0.70) and  $k=4$  (0.14), indicating the data support at most two broad groups. The held-out FMD comparison was stronger for K-Means in this rerun, while ultimately DBSCAN (discussed and shown above and below) was more useful for identifying unusual counties. These results are exploratory and do not imply causality.

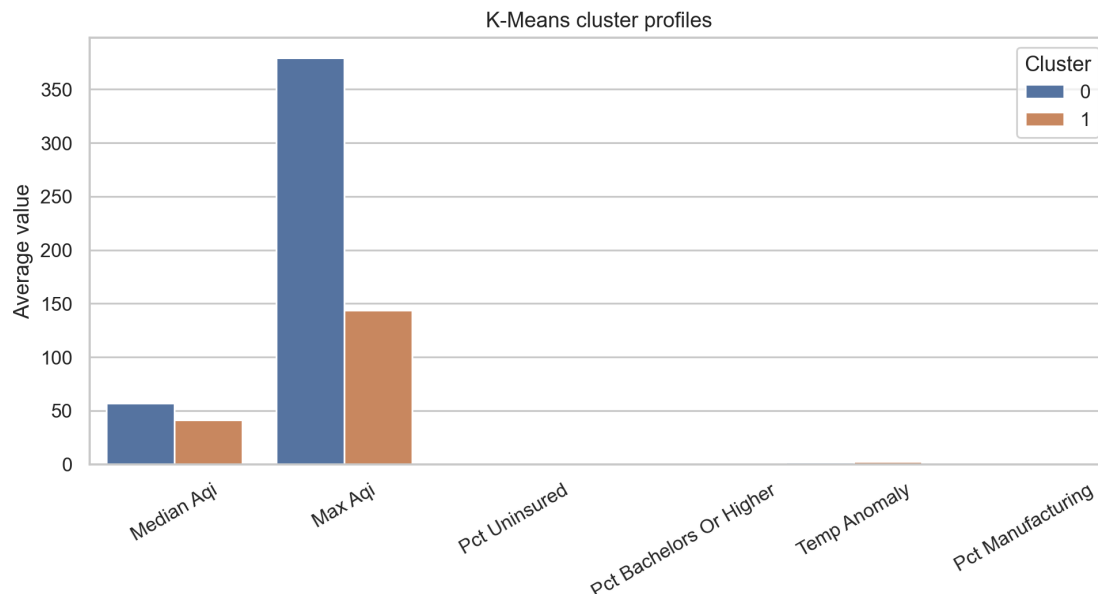


Figure 11. K-Means cluster profiles. Cluster 0 (27 counties) has extreme max AQI and lower FMD (ANOVA  $F=8.63$ ,  $p=0.003$ ).

## Discussion

The enriched feature set contains some predictive signal, but not enough for confident next-year FMD forecasting. The best model improved over the Dummy baseline and approached zero holdout  $R^2$ , but large residuals remained in counties with the highest observed mental distress. What surprised us most was that adding daily AQI features did not meaningfully improve prediction. To avoid data leakage, we predicted next-year FMD using only information available from the previous year. Leakage-derisking constrained the effective training window to only four year-pairs, making it substantially harder for the models to learn robust patterns. However, this tradeoff was necessary because it better reflects how the model would perform in a real forecasting setting. Our grouped ablation analysis showed that removing all 25 daily AQI variables slightly decreased RMSE ( $-0.004$ ), whereas removing education and health access variables produced the largest increase in RMSE ( $+0.08$ ). Given these results, it makes us wonder whether education and health access variables contributed more consistently to predictive performance than the annual and daily air quality feature families currently available.

Ultimately, the project's central empirical result is a negative finding: annual county-level environmental, climate, and socioeconomic variables did not provide strong next-year FMD prediction under a leakage-safe temporal holdout. We interpret this as evidence that the current feature set, spatial scale, and annual aggregation could benefit from improvement. Though we did not end up with the robust prediction model we might have optimistically liked, we believe this negative result still has value. It shows the limitations of annual county-level predictors while encouraging directions for future work:

1. reconstruct the outcome from raw BRFSS microdata,
2. incorporate monthly rather than annual exposure windows,
3. add healthcare provider density, RUCC rurality codes, disability prevalence, and other county-level health infrastructure variables, and
4. explore spatial and hierarchical regression models that better account for geographic dependence.

## Ethical Considerations and Notes of Caution

Because we had to impute a significant portion of the dataset, it is important to recognize that some county-level values are estimates rather than direct observations. Sparse monitor coverage, modeled PLACES estimates, and imputed environmental values all carry uncertainty that should be communicated and considered as a constraint. This introduces additional uncertainty into both the supervised and unsupervised analyses:

- County-level predictions should not be interpreted as individual-level mental health risk or used for diagnosis.
- High-FMD counties should not be stigmatized; maps should be used to guide support and further inquiry, not blame-directed speculation.
- Forecasting tools for mental health resource planning should be paired with local expertise and community input.
- Because the analysis is predictive and exploratory, it should not be used as causal evidence for regulatory decisions without further study.

Our unsupervised learning pipeline includes notes of caution:

- Clustering simplifies diverse counties into a small number of statistical groups, meaning local granularity is inevitably lost.
- The DBSCAN noise label should not be interpreted as meaning a county is abnormal or problematic; it simply highlights that the county did not belong to one of the dense groups identified under the selected clustering parameters.
- We view the emergent clusters as exactly that: an invitation to explore further, not explanatory for why communities differ.
- Because FMD was intentionally excluded from clustering and only compared across clusters afterward, we should be especially careful not to point blame or causality from the observed differences.

| Limitation                                      | Implication                                                  | Recommended next step                                                                        |
|-------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Annual county aggregation                       | Smooths short-term exposure events and local heterogeneity   | Use monthly or daily exposure windows and spatially finer outcomes where possible            |
| Sparse EPA daily AQI coverage                   | Daily AQI features are available only for monitored counties | Model monitor coverage explicitly or add satellite/reanalysis pollution estimates            |
| Negative income sentinel values in source panel | Requires cleaning before modeling and profile interpretation | Treat sentinel values as missing and rerun models                                            |
| Missing healthcare and rurality variables       | May explain residual patterns not captured by AQI/climate    | Add provider density, RUCC/NCHS urban-rural class, opioid burden, disability, and greenspace |
| No causal identification strategy               | Associations should not be framed as pollution effects       | Use quasi-experimental or longitudinal causal designs for policy claims                      |

*Table 9. Limitations and recommended next steps.*

## Statement of Work

| Team member      | Primary contributions                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Jaeah Kim        | Unsupervised learning analysis support, formatting, clustering design, interpretation support.                                          |
| Kyle Rodriguez   | Supervised and unsupervised evaluation, repository cleanup, data gathering, initial baselines for supervised learning report synthesis. |
| Sophia Boettcher | Dataset integration, imputation workflow, supervised and unsupervised baselines support, final report drafting support.                 |
| All team members | EDA, data cleaning, modeling review, interpretation, and final presentation/report synthesis.                                           |

*Table 10. Statement of work.*

## Appendix A - Links to Project & Bibliography

### Project Submission

GitHub Repository: <https://github.com/sboettch/SIADS696-Team7-Final>

Google Doc Report:

<https://docs.google.com/document/d/13ooJPQY85FIEtRn-MrTg2tTR2SjfYgQbQDaTFxOP0YY/edit>

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## Appendix B - Full Table of Features

| Feature family                  | Variable names                                                                                                                            | Data type             | Data source             | Notes                                                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geographic identifiers          | geoid, county_name, state_name, year                                                                                                      | Integer / Categorical | Census                  | County and temporal identifiers used for merging datasets.                                                                                                                                                                     |
| Mental health outcome           | mental_health_prevalence (Frequent Mental Distress, FMD)                                                                                  | numerical (float64)   | CDC PLACES              | Target variable for supervised learning and hold-out outcome for cluster evaluation. Frequent mental distress among adults; a.k.a rate wherein mental health is not good for $\geq 14$ days among adults aged $\geq 18$ years. |
| Air quality                     | median_aqi, max_aqi, ninety_percentile_aqi, days_with_aqi, days_pm25, days_pm10, days_ozone, days_no2, days_co, unhealthy_sensgroups_days | numerical (float64)   | EPA AirData             | Captures annual air quality burden, pollutant-specific exposure, and extreme pollution events.                                                                                                                                 |
| Engineered air-quality features | bad_air_day_ratio, good_air_day_ratio                                                                                                     | numerical (float64)   | Engineered from AirData | Summarizes overall exposure to unhealthy versus healthy air quality days.                                                                                                                                                      |
| Climate                         | Annual temperature, precipitation, climate anomalies                                                                                      | numerical (float64)   | NOAA                    | Provides environmental context beyond air pollution.                                                                                                                                                                           |
| Socioeconomic                   | median_income, population, poverty_rate, unemployed_rate                                                                                  | Integer / Float       | ACS                     | Represents county-level socioeconomic conditions associated with community vulnerability.                                                                                                                                      |
| Education & healthcare access   | Bachelor's degree attainment, uninsured rate                                                                                              | numerical (float64)   | ACS                     | Proxies for educational attainment, healthcare access, and community resources.                                                                                                                                                |
| Industry composition            | Manufacturing, agriculture/mining, education & health services, professional & administrative employment shares                           | numerical (float64)   | ACS                     | Characterizes county economic structure and supports interpretation of discovered clusters.                                                                                                                                    |